ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

# Geometric Sequences

*Geometric sequences multiply by one constant ratio.*

GEOMETRIC SEQUENCES · P2 · 06

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# The Map

This strategy booklet was mined from the local topic problem/answer PDFs, Dr Eslam's source notes, and the WMA12 website answer bank. The topic reduces to these exam moves, with hard variants kept visible on each page.

**01** RATIO: Common Ratio

TRIGGER *geometric sequence*

**02** TERM: Nth Geometric Term

TRIGGER *find* $u_n$

**03** SUM: Finite Geometric Sum

TRIGGER *sum of first n geometric terms*

**04** INFTY: Sum To Infinity

TRIGGER *infinite geometric series*

BANK EVIDENCE Local pages: 25 problem, 46 answer, 7 notes. Website primary entries: 7.

# 01 RATIO: Common Ratio

TRIGGER *geometric sequence*

BECOMES Divide consecutive terms.

FIRST LINE TO WRITE

$$r = \frac{u_{n+1}}{u_n}$$

SIMPLEST STRATEGY

1. **R**ead consecutive terms.
2. **A**rrange next over previous.
3. **T**ake the ratio.
4. **I**nsert signs/fractions.
5. **O**btain next terms.

WORKED MODEL

Sequence $3, 6, 12$ has $r = 2$.

HARD VARIANTS

1. If terms are not consecutive, use a power of $r$.
2. Negative ratios alternate signs.
3. Fractional ratios create decay.

BOTTOM LINE

Geometric means multiply by the same amount.

# Practice 01

SAME IDEA Use RATIO: Common Ratio.

QUESTION

Find $r$ for $81, 27, 9, \ldots$.

YOUR SOLUTION

# 02 TERM: Nth Geometric Term

TRIGGER *find $u_n$*

BECOMES Use first term times ratio to the $n-1$.

### FIRST LINE TO WRITE

$$u_n = ar^{n-1}$$

### SIMPLEST STRATEGY

1. **T**ake $a$.
2. **E**valuate $r$.
3. **R**aise to $n-1$.
4. **M**ultiply and simplify.

### WORKED MODEL

If $a = 5, r = 2$, $u_n = 5 \cdot 2^{n-1}$.

### HARD VARIANTS

1. For missing $n$, use logs if needed after forming $ar^{n-1} = \text{value}$.
2. Keep exact powers where possible.
3. Check whether $n$ must be an integer.

### BOTTOM LINE

The exponent counts jumps from the first term.

# Practice 02

SAME IDEA Use TERM: Nth Geometric Term.

QUESTION

> Find $u_8$ for $a = 3, r = \frac{1}{2}$.

YOUR SOLUTION

# 03 SUM: Finite Geometric Sum

TRIGGER *sum of first n geometric terms*

BECOMES Use the finite sum formula.

### FIRST LINE TO WRITE

$$S_n = \frac{a(1-r^n)}{1-r}$$

### SIMPLEST STRATEGY

1 **S**ecure $a, r, n$.

2 **U**se correct sign version.

3 **M**ake the substitution.

### WORKED MODEL

$$S_5 = \frac{3(1-2^5)}{1-2} = 93.$$

### HARD VARIANTS

1 If $r > 1$, the formula still works; signs cancel.

2 If $r = 1$, use $S_n = an$, not the formula.

3 If a sum is given, solve the resulting exponential equation carefully.

### BOTTOM LINE

Finite geometric sums depend on $r^n$.

# Practice 03

SAME IDEA Use SUM: Finite Geometric Sum.

QUESTION

Find $S_6$ for $a = 4, r = 3$.

YOUR SOLUTION

# 04 INFTY: Sum To Infinity

TRIGGER *infinite geometric series*

BECOMES Use the formula only when $|r| < 1$.

### FIRST LINE TO WRITE

$$S_\infty = \frac{a}{1-r}$$

### SIMPLEST STRATEGY

1. **I**nspect $|r|$.
2. **N**ote convergence.
3. **F**orm $a/(1-r)$.
4. **T**idy exact value.
5. **Y**ield answer.

### WORKED MODEL

If $a = 10, r = \frac{1}{2}, S_\infty = 20$.

### HARD VARIANTS

1. State the convergence condition before using the formula.
2. For word problems, interpret the limiting total.
3. Negative ratios can converge if their magnitude is less than 1.

### BOTTOM LINE

No convergence, no sum to infinity.

# Practice 04

SAME IDEA Use INFTY: Sum To Infinity.

QUESTION

> Find $S_\infty$ for $a = 6, r = -\frac{1}{3}$.

YOUR SOLUTION

# Quick Reference

TRIGGER → FIRST ACTION

| TRIGGER | FIRST ACTION |
|---|---|
| geometric sequence | $r = \dfrac{u_{n+1}}{u_n}$ |
| find $u_n$ | $u_n = ar^{n-1}$ |
| sum of first $n$ geometric terms | $S_n = \dfrac{a(1-r^n)}{1-r}$ |
| infinite geometric series | $S_\infty = \dfrac{a}{1-r}$ |